

Practice Intermediate Algebra Test I (ver 2)

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Intermediate Algebra).
- II) Show all work.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.

① Determine if the following system has one solution, no solution, or infinite solutions. Prove algebraically.

$$\begin{cases} y = -3x + 1 \\ y = 2x + 1 \end{cases}$$

② Simplify and write in $a+bi$ form.
 $(3i+4)^2$

③ Solve.

$$-3 < 4 + \frac{-5x+3}{2}$$

④ Rationalize the denominator and simplify.

$$\sqrt[4]{\frac{80}{3}}$$

⑤ Multiply and simplify.
Assume that x is positive.

$$(\sqrt{3x^7y})(\sqrt{6x^3y^4})$$

⑥ Simplify. ~~★~~ Do not
assume that the variables
are positive.

$$\sqrt{24x^6y^2}$$

⑦ Does the equation below have one solution, no solution, or infinite solutions? Prove algebraically.

$$2(x-1)+3=3(x-2)-x+8$$

⑧ Solve.

$$\begin{cases} 5x+2y+3z=2 \\ 2x-4y+2z=12 \\ 3x-y+4z=10 \end{cases}$$

⑨ Solve.

$$8|x| - 2 = 5$$

⑩ Simplify.
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⑪ Find the discriminant of the following equation and use it to determine the number of and type of solutions. Write "two real solutions," "one real solution," or "two complex solutions."

$$0 = x^2 - 2x + 1$$

Simplify and write in $a+bi$ form.

⑫ $(4+7i) + (-3-2i)$

⑬ $\frac{i-2}{i+3}$

⑭ Rationalize the denominator.

$$\frac{2}{\sqrt{5x}-3}$$

⑮ Solve by completing the square.

$$x^2 + 3x + 2 = 0$$

⑯ Solve.

$$\begin{cases} 4x - y + z = -11 \\ x + y - 2z = -7 \\ 2x + 9 = z \end{cases}$$

Solve problem #'s 17-19 by completing the square.

⑪ $x^2 - 6x - 16 = 0$

⑫ $5x^2 + 10x + 40 = 0$

⑬ $0 = 4x^2 - 5x - 3$

⑭ Simplify.

$$\frac{\sqrt{63}}{\sqrt{7}}$$

Solve using the quadratic formula.

(21) $2x^2 + 4x - 5 = 0$

(22) $0 = -3x^2 - 2x - 1$

Find the algebraic and graphical solutions.

(23) $|x - 2| < 10$

(24) $\frac{|x|}{3} - 2 > -2$

(25) Write in terms of i and simplify.

$3\sqrt{-12} + \sqrt{-48}$

Answers for Version 2

① One Solution

② $7 + 24i$

③ x can be any real number.

④ $\frac{2\sqrt[4]{135}}{3}$

⑤ $3x^5y^2\sqrt{2y}$

⑥ $2\sqrt{6}|x^3y|$

⑦ No Solutions

⑧ $(0, -2, 2)$

⑨ $x = -\frac{7}{8}$ or $\frac{7}{8}$

⑩ -1

⑪ $D=0$, One Real solution

⑫ $1 + 5i$

⑬ $-\frac{1}{2} + \frac{10}{2}i$

⑭ $\frac{2\sqrt{5x+6}}{5x-9}$

⑮ $x = -2$ or -1

⑯ $(-3, 2, 3)$

⑰ $x = 8$ or -2

⑱ $x = -1 \pm i\sqrt{7}$

⑲ $x = \frac{5 \pm \sqrt{73}}{8}$

⑳ 3

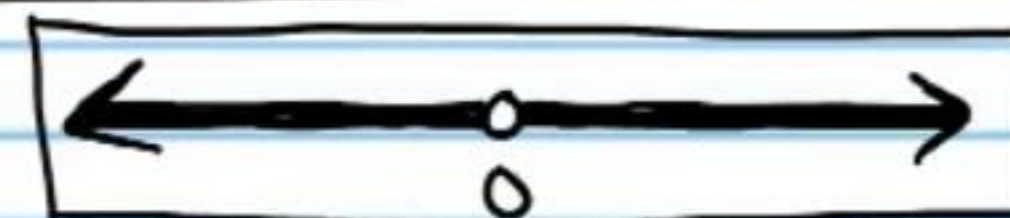
㉑ $x = \frac{-2 \pm \sqrt{14}}{2}$

㉒ $x = \frac{1 \pm i\sqrt{2}}{-3}$

㉓ $x < 12$ and $x > -8$



㉔ Any real number except zero.



㉕ $10i\sqrt{3}$